Sahaj Bhadja

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Summary

Software engineer with 2+ years building production systems and test workflows. Strong in **C++**, **Java**, **Python**, **Linux**, and debugging software.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Tech, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected **platform services** on **Azure SQL** and Blob Storage for 500+ users, optimizing queries to reduce dashboard latency from 1.2s to 350ms in production.
- Built fault-tolerant execution workflows with explicit state transitions and failure isolation for 1K+ jobs, improving throughput 3x under concurrent load and simplifying production debugging.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered **Flask** services and **REST APIs** for AI grading workflows serving 10K+ daily requests, integrating secure file ingestion, scoring pipelines, and instructor-facing features.
- Designed concurrent batch-processing with multithreading on **AWS RDS**, handling 1K+ file batches while maintaining stable APIs and reducing turnaround time by 60%.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built and evaluated spatial-temporal planning pipelines over 100K+ trajectory logs, improving closed-loop decision stability by 15% while strengthening reproducibility in simulation workflows.
- Developed experimental tooling for sensor-driven systems, using iterative validation and failure analysis to measure behavior and communicate findings across interdisciplinary collaborators.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered microservice-based software with **Jenkins**, **PyTest**, and **Mockito** across Windows, Linux, and macOS, improving release reliability through 20+ test suites.
- Optimized API and ETL workflows with **AWS Glue**, **S3**, and object-oriented design patterns, improving response times by 30% and pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed workflow software with **REST APIs**, **WebSockets**, and microservices, supporting 2M+ synchronized records while improving maintainability and real-time coordination.
- Automated Docker-based build and deployment workflows with **Jenkins CI/CD**, cutting release time by 50% and improving troubleshooting speed and environment consistency.

Skills

Languages: C++, Java, Python, SQL, JavaScript, TypeScript, Bash
Embedded & Systems: object-oriented design, algorithms, data structures, STL, multithreading, Linux, Windows, debugging
Backend & APIs: Flask, FastAPI, Node.js, REST APIs, microservices, WebSockets, SQL Server, PostgreSQL, MySQL, Azure SQL
Tools & Platforms: Git, Jenkins, Docker, AWS, Azure, GCP, S3, Blob Storage, CI/CD, unit testing, ETL pipelines

Projects

CPU-scheduling Simulator Live | *C++, React, EJ-2*
- Implemented FCFS, SJF, SRT, RR, priority scheduling, and MLQ with **C++** logic, metrics, and Gantt visualization across 1K+ simulated process combinations.

Chat2SQL Code | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-powered SQL workspace supporting 10+ databases, accelerating query authoring, schema exploration, and data management across a **Java** and TypeScript codebase.

HMIS Live | *MERN, TypeScript, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, Twilio alerts, Sentry monitoring, and scalable workflows to reduce administrative workload by 30%, emphasizing reliable operational logic and issue visibility.